Mid-Term Study Guide: Chapters 1–2

Linear and Quadratic Functions
Lessons 1-1 to 2-3: read, copy, practise, then sit the mock

How to use this guide

The exam: 55 minutes, 20 marks. Section A: 15 multiple-choice questions (0.5 mark each). Section B: 4 matching pairs. Section C: 4 fill-in-the-blank items. Then 2 short-answer questions and one problem-solving question with a table and a graph sketch.

Coverage: Chapter 1 Linear Functions (Lessons 1-1 to 1-6, about two-thirds of the paper) and Chapter 2 Quadratic Functions (Lessons 2-1 to 2-3, about one-third).

Plan: (1) read one lesson and copy each example without looking; (2) do the practice for that lesson; (3) after all lessons, sit the mock exam in 55 minutes with no notes; (4) mark it with the key and redo the examples for every lost mark.

Chapter 1: Linear Functions

Lesson 1-1: Key features of functions

Domain = all allowed inputs x; **range** = all outputs y. **x-intercept:** where $f(x) = 0$. **y-intercept:** $f(0)$. **Slope:** $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ is the rate of change. A function is **linear** when the slope is the same between every pair of points. **Slope-intercept form:** $f(x) = mx + b$. **Increasing** where the graph rises left to right, **decreasing** where it falls.

Example 1: intercepts of $f(x) = 3x - 12$

x-intercept: $3x - 12 = 0$ gives $x = 4$, so $(4, 0)$. y-intercept: $f(0) = -12$, so $(0, -12)$.

Example 2: is the table linear?

x	0	1	2	3
$f(x)$	5	3	1	-1

Each step of 1 in x changes f by -2, so the slope is constant: linear, with $f(x) = -2x + 5$. It is decreasing.

Example 3: slope through $(-1, 4)$ and $(3, -4)$

$$m = \frac{-4 - 4}{3 - (-1)} = \frac{-8}{4} = -2.$$

Trap: subtract the coordinates in the *same order* on top and bottom.

Example 4: intervals

A graph rises until the point $(2, 5)$ and then falls. It is increasing on $(-\infty, 2)$, decreasing on $(2, \infty)$, and its maximum value is 5 at $x = 2$.

Lesson 1-2: Transformations of functions

$f(x) + k$: up k. $f(x) - k$: down k. $f(x - h)$: **right** h. $f(x + h)$: **left** h. $-f(x)$: reflect in the x-axis. $f(-x)$: reflect in the y-axis. $a\,f(x)$: vertical stretch by a if $|a| > 1$, compression if $0 < |a| < 1$.

Example 5: where does a point go?

$(2, 5)$ is on f. For $g(x) = 2f(x - 1) + 3$: the x-value moves right 1, so $x = 3$; the y-value is stretched then shifted, $2(5) + 3 = 13$. The point is $(3, 13)$.

Trap: inside the bracket the shift has the *opposite* sign, and it changes x. Outside, it changes y.

Example 6: $g(x) = -f(x + 2)$ with $(4, -1)$ on f

$x + 2$ moves left 2: $x = 2$. The minus reflects in the x-axis: $y = 1$. The point is $(2, 1)$.

Lesson 1-3: Piecewise-defined functions

A piecewise function uses a different rule on each interval of x. To evaluate, choose the piece whose condition the input satisfies. At a boundary, the piece with $\leq$ or $\geq$ includes the point (closed circle); the other piece has an open circle.

Example 7: $f(x) = \begin{cases} x+2 & x < 1 \\ 3x-1 & x \geq 1 \end{cases}$

$f(-3) = -3+2 = -1$. $f(1) = 3(1)-1 = 2$ (the second rule, because $x \geq 1$). $f(4) = 3(4)-1 = 11$.

Trap: at $x = 1$ the first rule would give 3, but it applies only for $x < 1$.

Example 8: writing a rule from a situation

A Riyadh car park charges 8 SAR for the first 2 hours and 3 SAR for each extra hour. $C(h) = \begin{cases} 8 & 0 < h \leq 2 \\ 8+3(h-2) & h > 2 \end{cases}$. For 5 hours: $C(5) = 8+3(3) = 17$ SAR.

Lesson 1-4: Arithmetic sequences and series

Explicit rule: $a_n = a_1 + (n-1)d$, where d is the common difference. **Sum of n terms:** $S_n = \dfrac{n\,(a_1 + a_n)}{2}$.

Example 9: terms

For $5,\ 9,\ 13, \ldots$: $d = 4$, so $a_n = 5 + (n-1)4 = 4n+1$ and $a_{20} = 81$.
Is 100 a term? $5 + (n-1)4 = 100$ gives $n - 1 = 23.75$, not a whole number, so no.

Example 10: find a_1 and d from two terms

$a_3 = 7$ and $a_8 = 22$. Then $5d = 22 - 7 = 15$, so $d = 3$, and $a_1 = 7 - 2(3) = 1$.

Example 11: sums

For $2,\ 5,\ 8, \ldots$: $a_{10} = 2 + 9(3) = 29$, so $S_{10} = \dfrac{10(2+29)}{2} = 155$.
A stand has 20 seats in row 1 and 3 more in each next row. Row 12 has $20 + 11(3) = 53$ seats, so the 12 rows hold $\dfrac{12(20+53)}{2} = 438$ seats.

Trap: the exponent in the rule is $n-1$, and the sum has the $\dfrac{n}{2}$ factor. Forgetting either gives a wrong answer that looks reasonable.

Lesson 1-5: Solving equations and inequalities by graphing

To solve $f(x) = g(x)$, graph both and read the x-coordinate of each intersection. To solve $f(x) > g(x)$, find the x-values where the graph of f is *above* the graph of g. To solve $f(x) = 0$, read the x-intercepts.

Example 12: $2x - 1 = -x + 5$

The lines $y = 2x-1$ and $y = -x+5$ meet at $(2, 3)$. The solution is $x = 2$. Check: $2(2)-1 = 3$ and $-2+5 = 3$.

Trap: the solution is the x-coordinate 2, not the point $(2, 3)$ and not the y-value 3.

Example 13: $2x - 1 > -x + 5$

For $x > 2$ the line $y = 2x-1$ is above $y = -x+5$, so the solution is $x > 2$.

Lesson 1-6: Linear systems

Substitution: replace one variable. **Elimination:** add or subtract equations to remove a variable. One solution: lines cross. **No solution:** parallel lines (same slope, different intercept). **Infinitely many:** the same line.

Example 14: substitution

$y = 2x - 1$ and $3x + y = 14$. Then $3x + 2x - 1 = 14$, so $5x = 15$ and $x = 3$; $y = 2(3) - 1 = 5$. Solution $(3, 5)$.

Example 15: elimination

$2x + 3y = 12$ and $4x - 3y = 6$. Add: $6x = 18$, so $x = 3$. Then $6 + 3y = 12$ gives $y = 2$. Solution $(3, 2)$.

Example 16: classify

$y = 3x + 2$ and $6x - 2y = -4$. The second gives $-2y = -6x - 4$, so $y = 3x + 2$: the same line, infinitely many solutions. Compare $y = 3x + 2$ and $y = 3x - 5$: same slope, different intercepts, no solution.

Example 17: word problem

At a Jeddah bookshop, 3 notebooks and 2 pens cost 19 SAR, and 1 notebook and 2 pens cost 11 SAR. Subtract: $2n = 8$, so $n = 4$ SAR. Then $4 + 2p = 11$ gives $p = 3.5$ SAR.

Chapter 2: Quadratic Functions

Lesson 2-1: Vertex form

$f(x) = a(x - h)^2 + k$ has **vertex** (h, k) and axis of symmetry $x = h$. If $a > 0$ it opens up and k is the minimum value; if $a < 0$ it opens down and k is the maximum value. Larger $|a|$ makes the parabola narrower. The y-intercept is $f(0) = ah^2 + k$.

Example 18: read the features of $f(x) = 2(x + 3)^2 - 5$

Vertex $(-3, -5)$; opens up since $a = 2 > 0$; minimum value -5; axis $x = -3$; y-intercept $f(0) = 2(9) - 5 = 13$.
Trap: $x + 3$ means $h = -3$, because the form is $x - h$.

Example 19: write the equation from the vertex and a point

Vertex $(1, 4)$ and the point $(3, 0)$. Use $f(x) = a(x-1)^2+4$: $0 = a(2)^2+4$, so $a = -1$. Answer: $f(x) = -(x-1)^2+4$.

Example 20: transformations of $y = x^2$

$g(x) = -(x - 2)^2 + 1$: reflect in the x-axis, shift right 2, shift up 1. Its vertex is $(2, 1)$ and 1 is its maximum value.

Lesson 2-2: Standard form

$f(x) = ax^2 + bx + c$ has y-intercept $(0, c)$, axis of symmetry $x = -\dfrac{b}{2a}$, and vertex $\left(-\dfrac{b}{2a}, f\left(-\dfrac{b}{2a}\right)\right)$.

Example 21: $f(x) = x^2 - 8x + 3$

Axis: $x = -\dfrac{-8}{2(1)} = 4$. Vertex: $f(4) = 16 - 32 + 3 = -13$, so $(4, -13)$, the minimum. y-intercept $(0, 3)$.
Trap: $-b$ makes the sign flip: $b = -8$ gives $x = +4$.

Example 22: $f(x) = -2x^2 + 12x - 5$

Axis: $x = -\dfrac{12}{2(-2)} = 3$. $f(3) = -18 + 36 - 5 = 13$, so the maximum is 13 at $x = 3$.

Example 23: from vertex form to standard form

$f(x) = (x - 3)^2 + 2 = x^2 - 6x + 9 + 2 = x^2 - 6x + 11$.

Lesson 2-3: Factored form

$f(x) = a(x-p)(x-q)$ has zeros p and q (the x-intercepts), axis $x = \dfrac{p+q}{2}$, and y-intercept $a\,p\,q$. Zero-product property: if $ab = 0$ then $a = 0$ or $b = 0$. A repeated factor $(x-p)^2$ gives one zero where the graph touches the axis.

Example 24: $f(x) = (x+6)(x-1)$

Zeros -6 and 1. Axis $x = \dfrac{-6+1}{2} = -2.5$. Vertex: $f(-2.5) = (3.5)(-3.5) = -12.25$.

Example 25: build the function

Zeros -2 and 3, through $(0,-12)$. $f(x) = a(x+2)(x-3)$; $-12 = a(2)(-3)$, so $a = 2$: $f(x) = 2(x+2)(x-3)$.

Example 26: solve $x^2 + 2x = 24$

Move everything to one side: $x^2 + 2x - 24 = 0$. Factor: $(x+6)(x-4) = 0$. So $x = -6$ or $x = 4$.

Trap: never set the factors equal to 24. The right side must be 0 first.

Which form shows what?

Standard ax^2+bx+c	shows the y-intercept $(0,c)$	axis $x = -\frac{b}{2a}$
Vertex $a(x-h)^2+k$	shows the vertex (h,k)	max or min value k
Factored $a(x-p)(x-q)$	shows the zeros p and q	axis $x = \frac{p+q}{2}$

Practice (answers at the end)

Chapter 1

1. Find the x- and y-intercepts of $f(x) = -2x + 10$.
2. Find the slope through $(2,9)$ and $(6,-3)$.
3. $(3,-2)$ is on f. Where is the image on $g(x) = -3f(x+4) - 1$?
4. Evaluate $f(-2)$, $f(0)$ and $f(3)$ for $f(x) = \begin{cases} 2x+1 & x < 0 \\ x^2 - 1 & x \geq 0 \end{cases}$.
5. Find the 25th term of $12,\ 9,\ 6, \ldots$ and the sum of its first 25 terms.
6. An arithmetic sequence has $a_5 = 14$ and $a_{12} = 35$. Find a_1 and d.
7. Solve $3x - 2 = -x + 6$ by finding where the graphs meet, then solve $3x - 2 < -x + 6$.
8. Solve $x + 2y = 8$ and $3x - 2y = 0$.
9. Two lines have equations $y = -x + 4$ and $2x + 2y = 8$. How many solutions does the system have?

Chapter 2

10. State the vertex, direction of opening and extreme value of $f(x) = -3(x-4)^2 + 7$.
11. Write the parabola with vertex $(-2,1)$ through $(0,9)$ in vertex form.
12. Find the axis and vertex of $f(x) = x^2 + 6x - 1$.
13. A water jet in a Jeddah park has height $h(t) = -5t^2 + 20t$ metres after t seconds. Find the maximum height and when the water lands.
14. Write the quadratic with zeros 1 and 5 through $(0,10)$.
15. Solve $x^2 - 3x = 28$ by factoring.

Answers to practice

Ch 1. 1. $(5,0)$ and $(0,10)$. 2. $\dfrac{-3-9}{6-2}=-3$. 3. $x=3-4=-1$ and $y=-3(-2)-1=5$: $(-1,5)$. 4. $f(-2)=-3$, $f(0)=-1$, $f(3)=8$. 5. $d=-3$, $a_{25}=12-72=-60$, $S_{25}=\dfrac{25(12-60)}{2}=-600$. 6. $7d=21$, $d=3$, $a_1=14-12=2$. 7. $4x=8$, $x=2$; the inequality is $x<2$. 8. Add: $4x=8$, $x=2$; $y=3$. Solution $(2,3)$. 9. The second equation is $y=-x+4$: the same line, infinitely many solutions.

Ch 2. 10. Vertex $(4,7)$; opens down; maximum value 7. 11. $f(x)=a(x+2)^2+1$; $9=4a+1$, so $a=2$: $f(x)=2(x+2)^2+1$. 12. Axis $x=-3$; $f(-3)=9-18-1=-10$; vertex $(-3,-10)$. 13. Axis $t=-\dfrac{20}{2(-5)}=2$; $h(2)=-20+40=20$ m; lands when $-5t(t-4)=0$, so at $t=4$ s. 14. $f(x)=a(x-1)(x-5)$; $10=5a$, so $f(x)=2(x-1)(x-5)$. 15. $x^2-3x-28=(x-7)(x+4)=0$, so $x=7$ or $x=-4$.

GRADE 10 · ALGEBRA II · MID-TERM · MOCK EXAM · 20 MARKS · 55 MINUTES

Mid-Term Mock Exam: Chapters 1–2

Linear and Quadratic Functions
No notes. Calculator not needed.

Name ..... Class _____ Date _____

Section A: Multiple choice

15 questions, 0.5 mark each

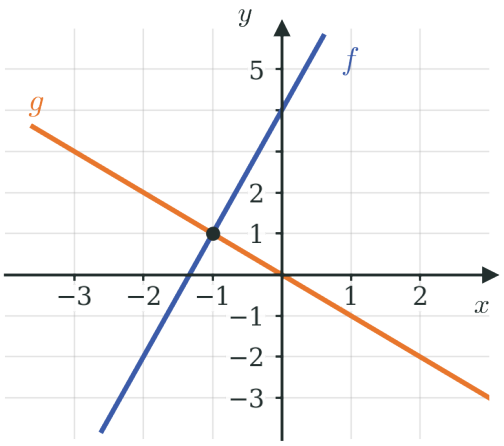
Circle ONE letter per question.

1 What is the x-intercept of the line $3x - 2y = 12$?

A. $(-4, 0)$ **B.** $(4, 0)$

C. $(0, -6)$ **D.** $(12, 0)$

2 The graphs of $y = f(x)$ and $y = g(x)$ are shown. Which gives the solution of $f(x) > g(x)$?



A. $x < -1$

B. $x > 1$

C. $x = -1$

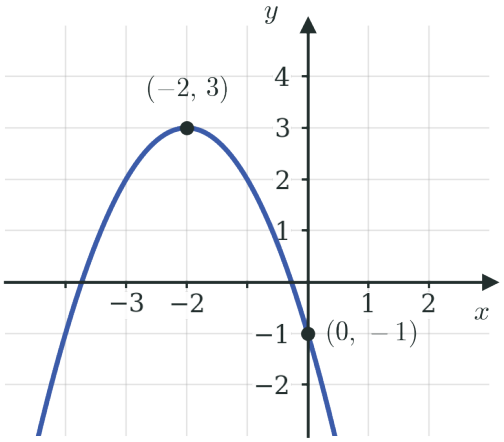
D. $x > -1$

3 How many terms are in the arithmetic sequence $5,\ 11,\ 17,\ \ldots,\ 83$?

A. 14 **B.** 13

C. 15 **D.** 78

4 The graph of a parabola is shown. Which equation matches it?



A. $f(x) = -(x-2)^2 + 3$

B. $f(x) = (x+2)^2 + 3$

C. $f(x) = -(x+2)^2 + 3$

D. $f(x) = -(x+2)^2 - 1$

5 Let $f(x) = \begin{cases} 3 - x & x \le 2 \\ 2x - 1 & x > 2 \end{cases}$. What is $f(2) + f(5)$?

A. 10 **B.** 12

C. -1 **D.** 9

6 The graph of f is reflected in the x-axis and then shifted 4 units to the left to form the graph of g. Which equation defines g?

A. $g(x) = -f(x-4)$ **B.** $g(x) = f(-x-4)$

C. $g(x) = -f(x+4)$ **D.** $g(x) = -f(x) + 4$

7 What is the axis of symmetry of $f(x) = (x+3)(x-7)$?

A. $x = -2$ **B.** $x = 4$

C. $x = 5$ **D.** $x = 2$

8 The table shows values of a linear function f. Which rule defines f?

x	1	2	3	4
$f(x)$	5	2	-1	-4

A. $f(x) = -3x + 5$ **B.** $f(x) = -3x + 8$

C. $f(x) = 3x + 8$ **D.** $f(x) = 8x - 3$

9 For which value of k does the system $2x + 3y = 6$ and $4x + 6y = k$ have infinitely many solutions?

A. $k = 6$ **B.** $k = 24$

C. $k = 12$ **D.** $k = 3$

10 What is the minimum value of $f(x) = 3x^2 + 12x - 5$?

A. -17 **B.** 31

C. -5 **D.** -25

11 An arithmetic sequence has $a_3 = 7$ and $a_9 = 19$. What is the sum of its first 15 terms?

A. 315 **B.** 255

C. 510 **D.** 180

12 Layla finds the slope of the line through $(-2, 5)$ and $(4, -7)$ like this: $m = \dfrac{-7-5}{-2-4} = \dfrac{-12}{-6} = 2$. Which statement is correct?

A. She inverted the formula; the slope is $-\dfrac{1}{2}$.

B. She made no mistake; the slope is 2.

C. She should have added the y-values; the slope is $-\dfrac{1}{3}$.

D. She subtracted the x-values in the opposite order to the y-values; the slope is -2.

13 What are the solutions of $x^2 - 4x = 21$?

A. $x = 7$ or $x = -3$ **B.** $x = -7$ or $x = 3$

C. $x = 21$ or $x = 25$ **D.** $x = 3$ or $x = 7$

14 A Jeddah school-bus service charges a one-time registration fee of 100 SAR plus 50 SAR per month (model values). A family has paid 850 SAR in total. For how many months have they used the service?

A. 19 **B.** 15

C. 17 **D.** 8

15 Which function has no zeros?

A. $f(x) = (x-3)^2$ **B.** $f(x) = -(x+1)^2 + 4$

C. $f(x) = 2(x-4)^2 + 1$ **D.** $f(x) = 3(x-2)^2 - 12$

Section B: Matching

4 pairs, 0.5 mark each

The point $(2, -3)$ lies on the graph of f. Match each function g with the image of that point on the graph of g. Write the letter on the line. One option is not used.

a. $g(x) = f(x-2) + 3$ _____

b. $g(x) = -f(x) + 2$ _____

c. $g(x) = f(-x) - 4$ _____

d. $g(x) = 3f(x+1)$ _____

A. $(1, -9)$

B. $(-2, -7)$

C. $(0, 0)$

D. $(2, 5)$

E. $(4, 0)$

Section C: Fill in the blank

4 blanks, 0.5 mark each

Complete each statement. Reason it out; no table or sketch is needed.

a. A parabola has axis of symmetry $x = 3$ and one zero at $x = -2$. Its other zero is $x =$ _____.

b. The sequence $40,\ 33,\ 26,\ \ldots$ first has a negative term at term number $n =$ _____.

c. The line through $(1, 4)$ that is parallel to $y = -3x + 2$ has y-intercept $(0,$ _____ $)$.

d. The vertex of $f(x) = x^2 - 6x + k$ lies on the x-axis when $k =$ _____.

Section D: Short answer

Show your working

1 An arithmetic sequence has $a_4 = 17$ and $a_9 = 42$.

(a) Find the common difference and write the explicit rule a_n.

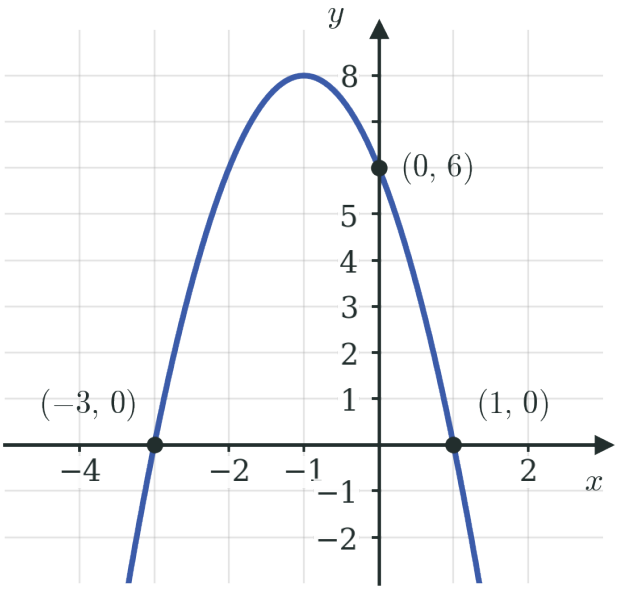
(b) Which term of the sequence equals 187?

(c) Hannah says that 100 is a term of the sequence. Is she correct? Show why. **[2 marks]**

2 The graph of a parabola $y = f(x)$ is shown. It crosses the x-axis at $(-3, 0)$ and $(1, 0)$ and the y-axis at $(0, 6)$. **[1.5 marks]**

(a) Write the equation of the parabola in factored form. **[1]**

(b) Write the function in the form $f(x) = ax^2 + bx + c$. **[0.5]**



Section E: Problem solving

5 marks

A Jeddah school runs a National Day fair and trials ticket prices. The table shows how many tickets n it sells at each ticket price p SAR (model values). The relationship is linear.

Ticket price p (SAR)	2	4	6	8
Tickets sold n	200	160	120	80

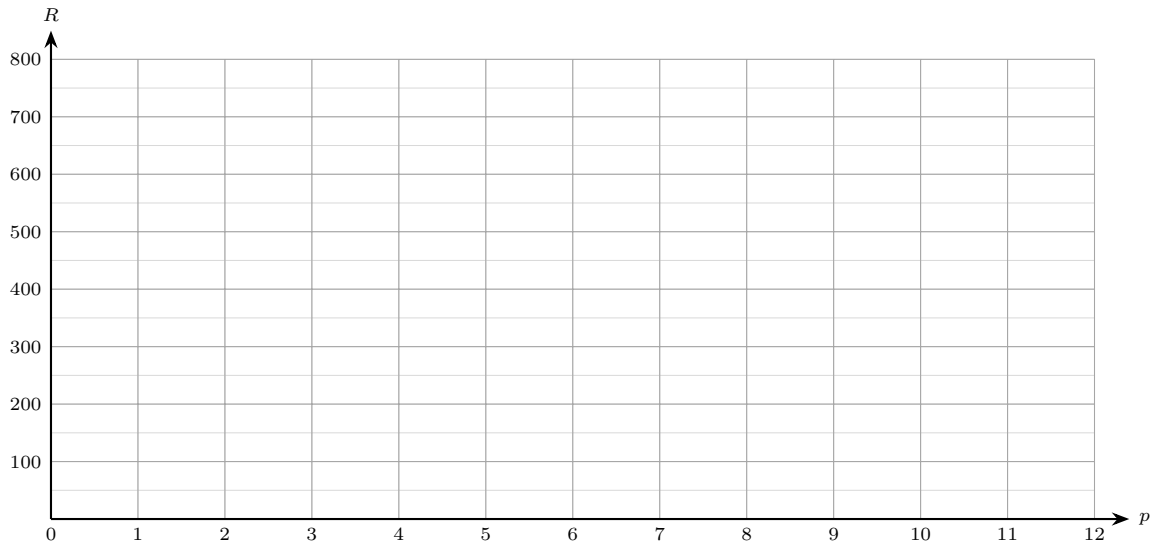
(a) Write a rule for n in terms of p. **[1 mark]**

(b) The revenue is $R(p) = p \times n$. Show that $R(p) = -20p(p - 12)$, and state the two prices at which the revenue is 0 SAR. **[1 mark]**

(c) Use symmetry to find the price that gives the greatest revenue, and that revenue. **[0.5 mark]**

(d) Complete the table, then plot the points and sketch the graph of R on the grid. Mark the vertex. **[1.5 marks]**

p (SAR)	0	2	4	6	8	10	12
$R(p)$ (SAR)	0			720			0



(e) The fair needs a revenue of at least 640 SAR. Draw the line $R = 640$ on your graph and state the prices that meet the target. Fatima says the school should choose the highest price that works. Give one reason, using the model, why a lower price in that range could be better. **[1 mark]**

Answer key and marking

Check your work
Mark each part, then revisit the matching example

Section A: multiple choice (0.5 mark each, 7.5 marks)

1 B **2** D **3** A **4** C **5** A **6** C **7** D **8** B **9** C **10** A **11** B **12** D **13** A **14** B **15** C

1 $3x = 12$ at $y = 0$. A: sign slip. C: the y-intercept. D: ignored the 3.

2 f is above g to the right of the crossing at $x = -1$. A: wrong side. B: used the y-value. C: solved $f = g$ only.

3 $\dfrac{83-5}{6} + 1 = 14$. B: forgot the $+1$. C: counted a term too many. D: stopped at $83 - 5$.

4 Vertex $(-2, 3)$, $a < 0$, $f(0) = -1$. A: sign of h. B: lost the minus in a. D: used the y-intercept for k.

5 $f(2) = 3 - 2 = 1$ (the $x \le 2$ rule), $f(5) = 9$; sum 10. B: used $2x - 1$ at $x = 2$. C: used $3 - x$ at $x = 5$. D: left out $f(2)$.

6 Reflect: $-f(x)$; 4 left: $x \to x + 4$. A: right shift. B: wrong axis. D: shifted up.

7 $\dfrac{-3+7}{2} = 2$. A: read zeros as $3, -7$. B: forgot to halve. C: half the distance between zeros.

8 $m = \dfrac{2-5}{2-1} = -3$; $f(0) = 8$. A: used $f(1)$ as b. C: sign of slope. D: swapped m, b.

9 Scale by 2: $k = 12$. A: left 6. B: scaled by 4. D: halved.

10 Axis $x = -\dfrac{12}{6} = -2$; $f(-2) = 12 - 24 - 5 = -17$. B: axis $+2$. C: the y-intercept. D: dropped the 3.

11 $d = \dfrac{19-7}{6} = 2$, $a_1 = 3$, $a_{15} = 31$, $S = \dfrac{15(34)}{2} = 255$. A: took a_3 as first term. C: no $\frac{1}{2}$. D: $d = \dfrac{12}{12}$.

12 Order matters: $\dfrac{-12}{6} = -2$. A: inverted. B: accepted the error. C: added y-values.

13 $(x - 7)(x + 3) = 0$. B: signs reversed. C: set factors equal to 21. D: both positive.

14 $100 + 50m = 850$, $m = 15$. A: $(850 + 100) \div 50$. C: ignored the fee. D: swapped fee and rate.

15 $2(x - 4)^2 + 1 \ge 1$. A: touches the axis (one zero). B: opens down, so it crosses $(x = 1, -3)$. D: zeros $0, 4$.

Sections B and C (0.5 mark per part, 4 marks)

B: a–E $(4, 0)$, b–D $(2, 5)$, c–B $(-2, -7)$, d–A $(1, -9)$; unused C is $f(x+2)+3$. **C:** a. 8 (mirror of -2 in $x = 3$). b. 7 ($a_n = 47 - 7n$, $a_7 = -2$, $a_6 = 5$). c. 7 ($y = -3x + 7$). d. 9 ($f(3) = 9 - 18 + k = 0$).

Section D: short answer (3.5 marks)

1(a) $5d = 42 - 17$, $d = 5$ [0.5]; $a_1 = 17 - 3(5) = 2$, $a_n = 5n - 3$ [0.5]. **(b)** $5n - 3 = 187$, $n = 38$ [0.5]. **(c)** $5n - 3 = 100$ gives $n = 20.6$, not a whole number, so 100 is not a term [0.5].

2(a) $f(x) = a(x + 3)(x - 1)$; $6 = a(3)(-1)$, so $a = -2$: $f(x) = -2(x + 3)(x - 1)$ [1]. **(b)** $-2(x^2 + 2x - 3) = -2x^2 - 4x + 6$ [0.5].

Section E: problem solving (5 marks)

(a) slope $= \dfrac{160 - 200}{4 - 2} = -20$; $n = -20p + 240$ [1]. **(b)** $R = p(240 - 20p) = -20p(p - 12)$ [0.5]; $R = 0$ at $p = 0$ and $p = 12$ [0.5]. **(c)** axis $p = \dfrac{0 + 12}{2} = 6$; $R(6) = 6(120) = 720$ SAR [0.5]. **(d)** $R(2) = 400$, $R(4) = 640$, $R(8) = 640$, $R(10) = 400$ [0.5]; points plotted [0.5]; smooth parabola, vertex $(6, 720)$ marked [0.5]. **(e)** the line $R = 640$ meets the curve at $p = 4$ and $p = 8$, so $4 \le p \le 8$ [0.5]. Same revenue at 4 SAR, but $n = 160$ tickets instead of 80: more visitors for the same money (or lower price is fairer to families) [0.5].

Scoring

Total 20 marks. 16 or more: ready. 12 to 15.5: redo the examples for every lost mark, then retry the section. Below 12: re-read the guide lesson by lesson before another attempt.